Introduction

- tidy data is tidy when...
  - each row = a unit of observation
  - each column = a measure on some variable
  - each entry contains a single data value
- data → large definition
  1. emails
  2. social media posts
  3. images
  4. audios
  5. videos

  } all count!
- data science has lots of uses!
- workflow for class: data visualization → data analysis → data storytelling
- software = R + RStudio

RStudio + HW1

- Console only good for
  1. quick calculations
  2. trying out code
  3. Pulling up help files

  → does not communicate w/ Quarto!
- Quarto = a technology that incorporates code from many programming languages
- Use "#" to leave short notes
- we will use a simplified version of the tidyverse style guide (GDoc)

Univariate Viz

- Visualizations = important for telling a story
- Data viz is the process of mapping data to different plots

- Graph Components
  1. a frame, or coordinate system
  2. a layer
  3. scales
  4. Faceting
  5. a them
- What we use to build (most) viz: ggplot
- tidyverse package: compiles a set of ind. packages that share a common structure (ex: ggplot)

Multivariate

- Defining fill or color depends on we define it by a named color or some variable
- Not often one "best plot"
  → each offer + loose things
- Scales: change color, fill, size, or shape

Spatial Viz.

Simpson paradox ⟶ confounders

Point Map → plotting locations of individual observations

Contour Map → plotting density/distribution of observations

Chloropleth Maps → plotting outcomes in different regions

Part 1= datasite; Part 2= background map

Leaflet = open-source that creates maps

Effective Viz

- Benefits ⟶ understand data set, refine research questions, communicate findings/tell a story
- Ineffective viz categories
  1. Wrong
  2. Bad
  3. Ugly
- Vis needs
  1. axis labels
  2. figure caption
  3. "alt text" → text that can be read by a screen reader
  4. Color palette that is color blind friendly
  5. To be ethical
     - visible/credit
     - privacy
     - power (challenge)
     - emotional (be a person)
     - pluralism
     - contextual

Wrangling

Goals:

- keep only certain observations
- define new variables
- reformat / clean existing variables
- combine various datasets
- process "string" or text data
- explore + summarize various characteristics of the variables in the dataset

object |> function

- object on left is passed to the function on the right.

Dates

%in% can be used to list multiple values

!= not equal to

== equal to

> greater than

>= greater than or equal to